Is it a boy or a girl?

All organisms are made of cells, and each type of organism has a particular number of chromosomes. For example, each body cell of a dog has 28 chromosomes, while each body cell of a horse has 64 chromosomes. Human cells contain 46 chromosomes, or 23 pairs of chromosomes, or 2 sets of 23 chromosomes (= 2*n*).

Reproductive cells – sperm in males and eggs (ova) in females – always contain half the number of chromosomes of the body cells; that is, one set of 23 chromosomes (= 1*n*). This ensures that the baby also has 46 chromosomes, not 92. When an egg and sperm join during fertilisation, 23 chromosomes come from the mother and 23 come from the father. The fertilised egg has a complete set of 46 chromosomes; that is, 23 pairs of chromosomes.

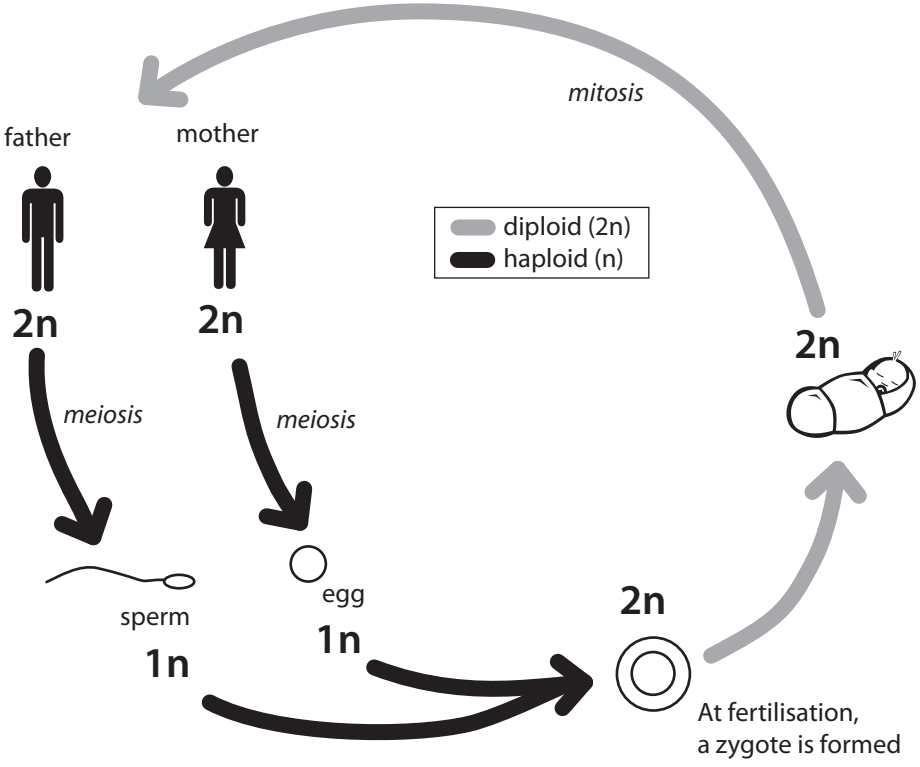


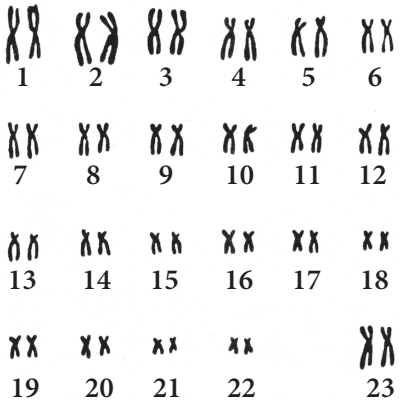
Figure 1

1 What do you think would happen if the egg and sperm always had a complete set of chromosomes?

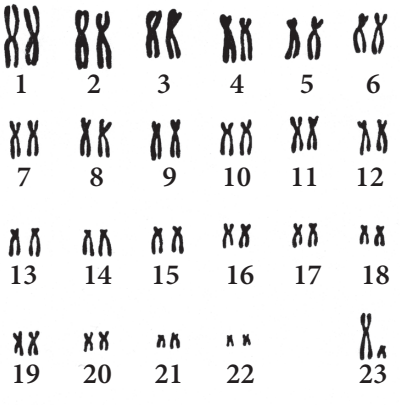
2 Find the terms 'diploid' and 'haploid' in figure 1. What do you think these terms mean?

A special pair of chromosomes determines whether the fertilised egg will be a boy or a girl. This pair is made up of the sex chromosomes. In males, one chromosome in this pair is smaller. The larger chromosome is called X and the smaller is called Y. Females have two X chromosomes, which are the same size. Males have X and Y. These X and Y chromosomes are called sex chromosomes, because they contain the genes that determine sex. The other chromosomes are called autosomes, and these include all the other pairs of chromosomes. Each pair of autosomes in the human karyotype is numbered.

Figure 2 shows the karyotype of a human female and a human male. The first 22 pairs of chromosomes are the autosomes. The 23rd pairs are the sex chromosomes.



Karyotype of a human female



Karyotype of a human male

Figure 2

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10b

As shown in figure 3, when a sperm with an X chromosome joins with an egg, the child will be a girl (XX). When a sperm with a Y chromosome joins with an egg, the child will be a boy (XY).

Males produce equal numbers of X and Y sperm, which means the chance of producing a boy or a girl is 50:50.

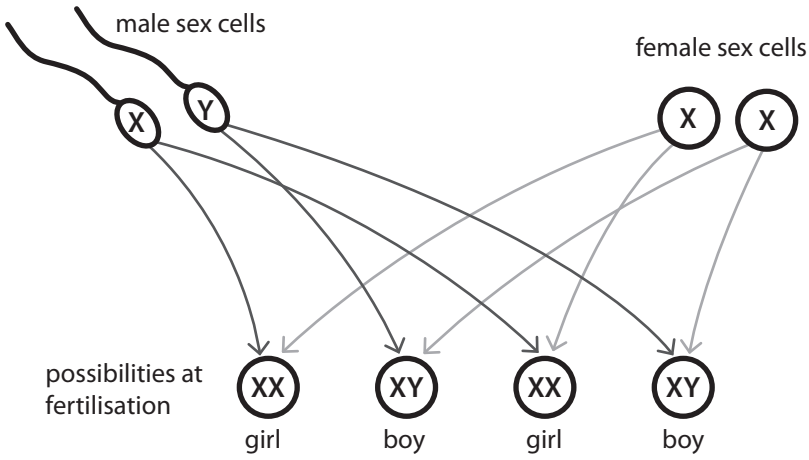


Figure 3

3 Define these terms.

a Autosome **b** Karyotype

4 How many chromosomes do humans have in:

a skin cells? **b** liver cells? **c** sperm?
d ovum? **e** nerve cells?

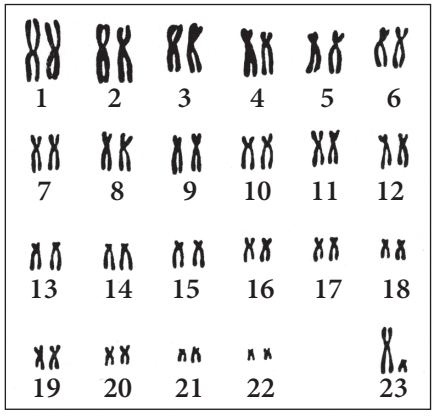
5 Describe how the sex of a human baby is determined.

6 Rewrite the summary points in the table in sentence form.

Location	Cell division	Description	Purpose
Somatic (body) cells	Mitosis	$2n$ (diploid) → $2n$ (diploid)	Growth and repair
Sex organs	Meiosis	$2n$ (diploid) → n (haploid)	Gamete production

7 A karyotype of a human baby is shown at the right.

a What sex is the baby? Explain how you know.
b How many chromosomes does the baby have?
c How many chromosomes did the mother pass on to baby?
d How many chromosomes did the father pass on to baby?
e How many chromosomes are there in the mother's skin cells?
f How many chromosomes are there in the mother's eggs?
g How many chromosomes are there in the father's sperm?
h How many chromosomes are there in the father's skin cells?



Karyotype of a human baby

8 **a** King Henry VIII reigned in England from 1509 to 1547 and had six wives. Historians believe that Henry beheaded his wives when they delivered baby girls, because he wanted a son to continue the royal succession. He blamed his wives for producing daughters instead of sons. Was Henry correct? Who determines the sex of a baby? Explain your answer.

b Why do you think this information was not known at the time of Henry VIII?

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